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Review article

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Transcriptomics in the era of long-read sequencing

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Abstract

Transcriptome sequencing revolutionized the analysis of gene expression, providing an unbiased approach to gene detection and quantification that enabled the discovery of novel isoforms, alternative splicing events and fusion transcripts. However, although short-read sequencing technologies have surpassed the limited dynamic range of previous technologies such as microarrays, they have limitations, for example, in resolving full-length transcripts and complex isoforms. Over the past 5 years, long-read sequencing technologies have matured considerably, with improvements in instrumentation and analytical methods, enabling their application to RNA sequencing (RNA-seq). Benchmarking studies are beginning to identify the strengths and limitations of long-read RNA-seq, although there remains a need for comprehensive resources to guide newcomers through the intricacies of this approach. In this Review, we provide a comprehensive overview of the long-read RNA-seq workflow, from library preparation and sequencing challenges to core data processing, downstream analyses and emerging developments. We present an extensive inventory of experimental and analytical methods and discuss current challenges and prospects.

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Introduction

Studying the transcriptome — the collection of RNA transcripts expressed under specific conditions — is instrumental to characterize cellular responses, infer gene regulatory networks, unveil disease mechanisms and discover diagnostic biomarkers[1]. For the past 20 years, short-read RNA sequencing (srRNA-seq) has been the gold standard for transcriptomics. However, srRNA-seq falls short of fully capturing the complexity and dynamics of the transcriptome[2]. Despite its widespread utility, srRNA-seq is constrained by the resulting 100–150 bp read length, which can reach ~300 bp in paired-end sequencing[3], making it difficult to resolve the connectivity between distant exons, as they are never represented on the same sequenced fragment[4]. Moreover, srRNA-seq cannot directly detect epitranscriptomic modifications and struggles with mapping repetitive regions, which limits its capacity to distinguish between similar transcripts. As a result, it is particularly challenging to study highly polymorphic genes, for example, HLA genes[5], and recombinant genes, such as immunoglobulin genes[6].

By contrast, long-read sequencing (LRS) methods such as nanopore sequencing, developed by Oxford Nanopore Technologies (ONT), and single-molecule real-time (SMRT) sequencing, developed by Pacific Biosciences (PacBio) (Fig. 1a), sequence entire RNA or cDNA molecules, which enables the recovery of full-length transcripts without assembly steps[7–9]. Features that distinguish very similar transcripts, such as differences at start and polyadenylation (poly(A)) sites, alternative exons, small in-phase nucleotide variants or modifications of the RNA molecule, can thus be captured in the same read (Fig. 1b). Consequently, long-read RNA sequencing (lrRNA-seq) has revealed new insights into transcriptome biology across diverse contexts. Illustrative examples include the identification of alternative splicing events in different brain regions and their association with neuropsychiatric and neurodegenerative diseases[10,11], gene fusions in cancer cell lines[12], alternative transcription start sites (TSSs) in thymus development, cell-state transitions and different cancer types and stages[13–15], and alternative transcription termination sites (TTSs) in neurodevelopment and evolution[16,17]. Additionally, LRS has driven the discovery of thousands of novel isoforms, even in well-annotated organisms[18,19], and can be used to study allele-specific expression[20] and RNA modifications[21,22], among other applications.

Initially, LRS technologies were limited by high sequencing error rates and low throughput[23,24]. However, improvements of platforms and library preparation methods over the past 5 years have enhanced accuracy, with tens of millions of reads generated in a single experiment[25,26]. The resulting increase in sample sizes has enabled the investigation of differential expression of full-length transcript models[14,27], profiling of genotypes[18,28], analyses in single cells[11,29] and spatial isoform expression[10,30]. Nowadays, there are algorithms for transcript reconstruction[31–33], quantification[34,35], differential expression[36,37], RNA modification detection[21,22], as well as advanced library preparation methods[25,38]. Additionally, benchmarking efforts have provided a comparative analysis of these experimental and computational methods[39,40], enhancing our understanding of their strengths and weaknesses (Box 1).

As LRS becomes more popular and accessible, new users face challenges in navigating the complexities of these technologies. This Review showcases the potential of lrRNA-seq and provides essential guidance for navigating both experimental and computational workflows. We provide a comprehensive overview of lrRNA-seq techniques, covering experimental design, library preparation, data processing, transcript detection, quantification and downstream analyses (Fig. 1c). We also address current challenges and future directions, aiming to help researchers effectively use LRS technologies for transcriptomic studies.

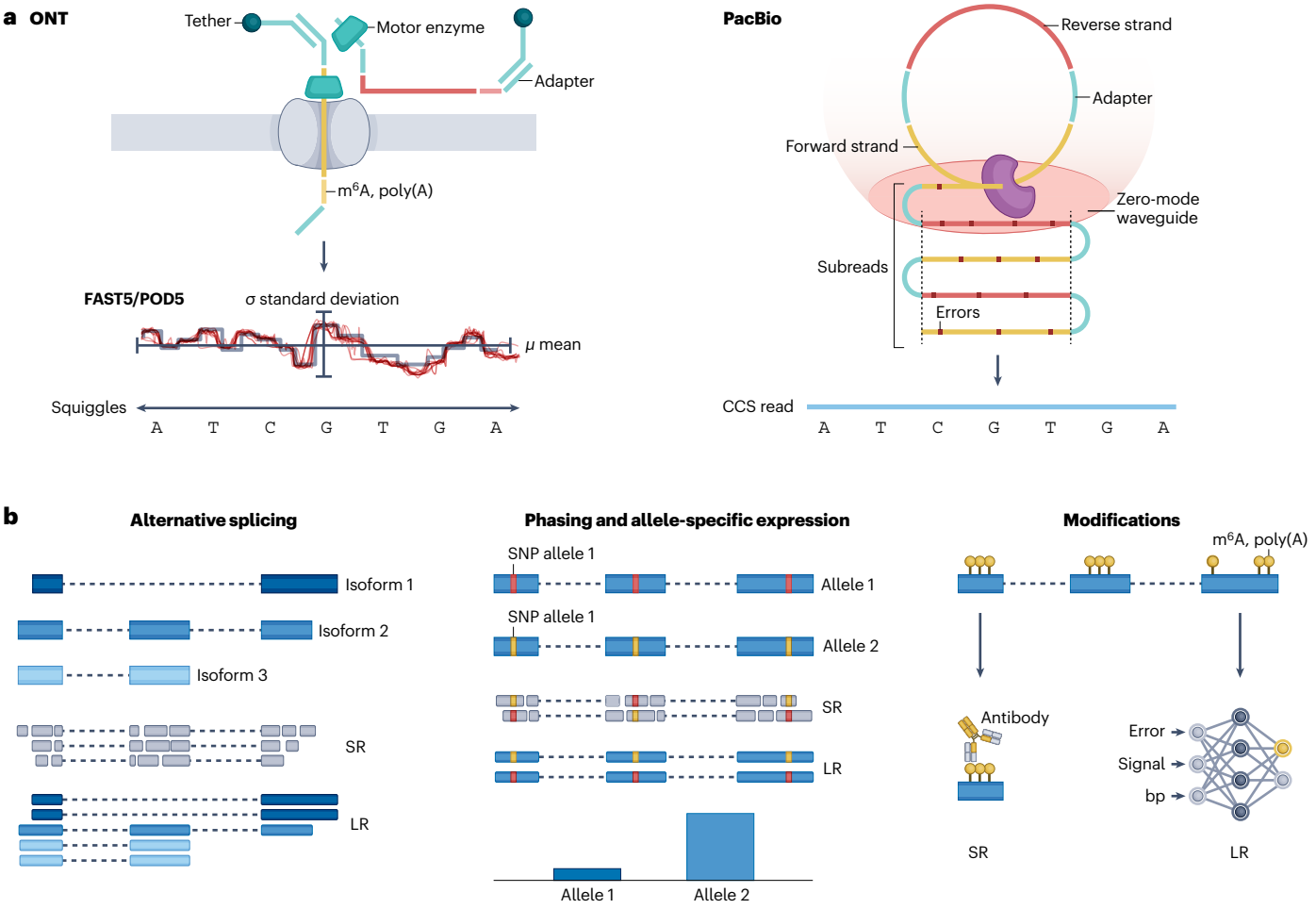
Experimental and sequencing design

Designing a robust experiment is critical to effectively answer biological questions. In lrRNA-seq, the factors to consider are RNA integrity, choice of sequencing platform, library preparation protocol, sequencing depth, multiplexing strategies and inclusion of external controls (Fig. 1c).

RNA quality

To effectively sequence full-length transcripts, high-quality RNA is essential. Degraded samples can produce incomplete transcript

Fig. 1 | Overview of long-read transcriptomics, applications and computational analyses. a, In long-read sequencing using Oxford Nanopore Technologies (ONT), cDNA or RNA molecules (in red and yellow) are tagged with sequencing adapters (in light blue) preloaded with a motor protein, and combined with a tether that localizes the molecules close to the flow cell. The sequencing occurs by introducing the sequencing adapter into a nanoscale pore and the motor protein unwinding the double-stranded molecules to drive them through the nanopore. As the molecule passes through the pore, the disruptions on the electric signal crossing the pore (squiggles) are measured and recorded in FAST5 or POD5 formatted files. Changes in current within the pore are used to identify the molecule sequences. In long-read single-molecule, real-time sequencing (SMRT) using Pacific Biosciences (PacBio), cDNA molecules (forward strand in yellow and reverse strand in red) are ligated to hairpin adapters (in light blue) to form circular molecules known as SMRTbells. SMRTbells are bound to polymerases (in purple) and immobilized in zero-mode waveguides. The polymerase synthesizes a new strand of cDNA using fluorescently labelled deoxynucleoside triphosphates (dNTPs) that are excited by light pulses. The fluorescent wavelengths are recorded, and the fluorophores are cleaved from the nucleotides to prevent fluorescent interference during the subsequent dNTP incorporation and light pulse. Multiple sequencing passes of the circularized cDNA molecule (subreads) allow for the generation of a consensus sequence of high accuracy. **b**, Core applications of long-read sequencing. Unlike short-read sequencing, long-read sequencing can capture full-length transcript isoforms in a single read, preserving exon continuity and identifying phased polymorphisms, which allows for allele differentiation and expression quantification and resolves the expression of highly similar transcripts. Additionally, although short-read sequencing of RNA modifications could only be performed indirectly by using RNA modification-specific antibodies to enrich fragments, or using chemical treatments to alter nucleotide bases, ONT directly detects epitranscriptomic modifications in the RNA molecule using unsupervised, supervised or semisupervised methods to identify specific disruption signatures in the electric current. **c**, The long-read RNA sequencing (lrRNA-seq) workflow: adequate experimental and sequencing design is necessary to obtain a data set that effectively addresses the biological questions of interest. The key steps for experimental and sequencing design, core data processing, downstream analysis and emerging applications are listed above the lines. The key considerations of each step are listed below the lines. Key factors include RNA quality, technology selection, library preparation methods, sequencing depth and the use of multiplexing and internal controls. Core data processing involves steps for transcriptome profiling and quality control. Downstream analyses encompass genome annotation, differential gene expression and functional interpretation. Emerging lrRNA-seq applications are expanding long-read sequencing to single-cell resolution and enabling the exploration of other aspects of RNA biology. AI, artificial intelligence; bp, base pairs; CCS, circular consensus sequencing; GC, guanosine–cytosine; GPU, graphical processing unit; LR, long read; poly(A), polyadenylation; scIso-Seq, single-cell RNA isoform sequencing; SNP, single nucleotide polymorphism; SR, short read.


a ONT
Tether
Motor enzyme
Adapter
m6A, poly(A)
FAST5/POD5
σ standard deviation
μ mean
Squiggles
A T C G T G A
PacBio
Reverse strand
Adapter
Forward strand
Zero-mode waveguide
Subreads
Errors
CCS read
A T C G T G A
b
Alternative splicing
Isoform 1
Isoform 2
Isoform 3
SR
LR
Phasing and allele-specific expression
SNP allele 1
Allele 1
SNP allele 1
Allele 2
SR
LR
Allele 1
Allele 2
Modifications
m6A, poly(A)
Antibody
SR
Error
Signal
bp
LR
c
Experimental design
Sequencing platform
Synthesis vs potential difference
Library type
cDNA vs direct RNA
Sequencing length
Single-molecule full-length
Sequencing depth
Power analysis
Sequencing design
Library preparation
Multiplexing, size selection
Quality control of detection, quantification
Spike-ins
Core data processing
Basecalling
AI algorithms, GPU, parallelization
Raw reads
Sequence quality, GC content, adapters, error correction
Read alignment
Complexity, alternative splicing
Transcript identification
Genome-free, reference-free, reference-supported
Transcript quantification
Only long reads, supported by short reads
Quality control
Of the data, transcriptome, tools
Downstream analysis
Differential expression
Isoform usage, alternative splicing
Annotation
Genomes, transcripts, functions
Functional analyses
Isoform co-expression, enrichment
Allele-specific expression
Phasing
Visualization
Genome browser
RNA modification
m6A, poly(A)
Emerging lrRNA-seq
Single-cell long reads
10×, scIso-Seq
Spatial long reads
Visium
Nascent RNA
Transcript regulation

Box 1 | Major take-home messages from benchmark studies

Benchmarking long-read RNA sequencing (lrRNA-seq) experiments have compared Oxford Nanopore Technologies (ONT) and Pacific Biosciences (PacBio) sequencing platforms, library preparation methods and transcript reconstruction pipelines, examining sequenced cell lines, spike-ins and simulated reads.

Challenges in transcript identification include differences in length distribution and read quality, influenced by sequencing platforms and library types[8,9,40]. Degraded RNA can lead to 5′ incomplete transcript model predictions, negatively impacting mapping accuracy and transcript quantification[8,39,62]. Mis-mapping of reads with sequence errors around splice junctions can result in transcript models with incorrect splice junction predictions[62]. Even the best-performing transcript identification tools have demonstrated low precision rates, highlighting the need for further method development to improve accuracy and encourage orthogonal validation of novel transcripts[40,70]. Replicability is particularly challenging for transcripts with medium-to-low expression levels[40,62]. Additionally, methods for sequencing library preparation and data processing need further refinement to distinguish expressed genes and stochastic noise in splicing machinery from sequencing noise[40,62].

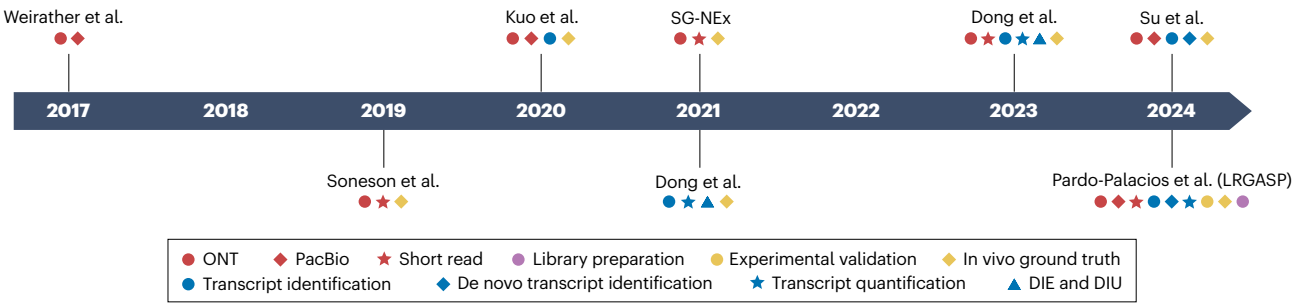
lrRNA-seq provides reduced ambiguity and greater power for differential transcript expression analysis compared with short-read RNA sequencing (srRNA-seq) for an equivalent number of reads[70]. The technology is also highly consistent, and both lrRNA-seq and srRNA-seq yield comparable gene expression estimates[49]. However, transcript reconstruction generates more accurate models from high-precision PacBio data sets than ONT[39,40,49]. Although increasing sequencing depth does not necessarily improve transcript model accuracy, it enhances sensitivity, enabling the detection of low-expression transcripts[39].

Long-read sequencing analysis uses tools that allow algorithmic tuning to accommodate sequencing error and biological assumptions[39,40,62]. Mapping methods with reference annotations increase the detection of true positives when high-quality reference annotations are available[39]. IsoQuant, StringTie2, Bambu and FLAIR are effective tools for isoform detection[39,40], with StringTie2 showing higher accuracy and Bambu greater power in detecting novel isoforms[70]. These tools, along with IsoQuant and FLAIR, have a high quantification correlation with srRNA-seq[40,70]. If the objective is to detect lowly expressed or rare transcripts, it is recommended to use tools supporting novel transcript detection and incorporating orthogonal data. Mandalorion and FLAIR offer balanced performance in terms of sensitivity and precision. For identifying a conservative set of highly supported novel transcripts, LyRic is an effective tool, although experimental validation of rare transcripts is still encouraged[40].

Although ONT direct lrRNA-seq is promising for complex transcriptome characterization and can produce transcript abundance estimates comparable to cDNA sequencing[49], it faces limitations, including low sequencing throughput, challenges in characterizing degraded or actively transcribing RNA and high error rates in terms of basecalled insertions and deletions[8,40,70], although overall read accuracy has improved with the newly released RNA004 chemistry[232].

Additionally, many benchmarking studies have generated big data resources that can be very useful in themselves to study different biases and specific aspects of long-read technologies, especially for the bioinformatics software development community. Some of these resources are the LRGASP[40], GTEx[18], ENCODE[233] and SG-NEx[49] (see the figure).

Despite these advances, lrRNA-seq chemistries and technologies are evolving faster than benchmarking efforts are being published. New independent benchmarking studies are needed to fully evaluate novel chemistries such as ONT's RNA004 and PacBio's SPRQ and Kinnex, as well as new flow cells such as ONT's R10 and emerging long-read sequencing technologies such as MGI's CycloneSeq. DIE, differential isoform expression; DIU, differential isoform usage.



sequences, making it difficult to differentiate alternative TSS and TTS and misrepresenting the complexity of the transcriptome. RNA degradation often results in biased coverage, with a tendency to over-represent the more stable 3′ ends of transcripts, leading to inaccurate isoform quantification. Additionally, degraded RNA compromises the ability to measure poly(A) tail lengths and introduces ambiguity in spliced alignment owing to shorter read lengths[41–43]. The RNA integrity number (RIN) measures the quality of RNA in a sample by analysing the ratio of intact ribosomal RNA peaks to degraded RNA fragments from an electropherogram; higher values indicate better RNA preservation[44]. Samples with a RIN > 9.5 in ONT sequencing were found to be undegraded, although samples with a RIN > 7 could be used for lrRNA-seq, provided that correction steps, such as NanoCount filtering according to best alignment per read, were implemented[45].

Alternatively, using orthogonal data to validate TSS and TTS[46], or not considering isoforms with alternative TSS and TTS in a differential isoform usage analysis by using transcript algorithms heavily reliant on the reference genome[32,33], could mitigate the biases of degraded RNA TSS and TTS in the lrRNA-seq analysis. Incorporating RIN as a covariate in statistical models aimed at identifying differentially expressed transcripts between conditions can account for degradation-related confounders[43]. Both ONT and PacBio recommend using RNA with a RIN > 8 to ensure accurate transcriptome representation[47,48]. Therefore, it is crucial to implement specific protocols for sample storage and RNA extraction that preserve RNA integrity as much as possible, tailored to the tissue or cell type under study.

Sequencing platform and library preparation

When selecting a sequencing platform, read length and accuracy are crucial (Fig. 1c and Table 1). LRS methods offer full-length molecule sequencing, but there are marked differences in both length distributions and read quality[9,40]. Both platforms can produce highly accurate sequences and capture full-length transcripts but have limitations to span the full range of lengths in the transcriptome. PacBio's commercial, accessible protocols are generally preferred when read accuracy and broad length range are priorities[39,40,49], whereas ONT's protocols are usually preferred when the priority is studying RNA modifications simultaneously with full-length transcripts[50–52].

ONT processes single DNA or RNA molecules in real time as they pass through a nanoscale pore embedded in a polymer membrane and disrupt an ionic current in ways indicative of their molecular sequence, using a motor protein to regulate the passage speed[26] (Fig. 1a). PacBio uses fluorescence signals for basecalling but differs from short-read sequencing methods by its continuous DNA synthesis. It uses zero-mode waveguides (ZMWs), small wells with a polymerase at the bottom, to confine the observation volume to a single circular cDNA molecule, novel phospho-linked nucleotides to ensure high accuracy and multiple-pass sequencing of the circular cDNA molecule to grant high fidelity. Engineered DNA polymerases synthesize DNA without interruption, producing reads that span thousands of bases[53] (Fig. 1a). Owing to these differences in chemistry, PacBio typically produces cDNA reads with higher accuracy than ONT[40,54]. Generally, ONT protocols can potentially sequence longer transcripts but have a bias towards selecting short fragments (Table 1) and have historically struggled with read accuracy[26,55]. However, read quality is strongly influenced by the library preparation method, which is platform-specific (Fig. 2).

Library preparation methods for ONT and PacBio usually begin with cDNA synthesis from poly(A)-selected RNA through an initial reverse transcription (RT) step, template switching and a second synthesis step to obtain double-stranded cDNA (Fig. 2a). After synthesis, ONT ligates a tether to the cDNA, which is then targeted and bound to the nanopore for sequencing[26] (Fig. 2b). By contrast, PacBio ligates universal hairpin adapters to the double-stranded cDNA fragments to create single-stranded circular cDNAs (to generate a so-called SMRTbell library) and anneals the sequencing primer to the hairpin adapters for sequencing in ZMWs[53] (Fig. 2c). Although PacBio sequencing produces high fidelity transcriptomes (>99% accuracy)[53], its lrRNA-seq library preparation protocols, Iso-Seq and Kinnex, include a size selection step that uses magnetic purification beads to generate the SMRTbell templates, which yield a consensus sequence from multiple sequencing passes; this size selection results in a selection bias towards a specific molecule length[9] (Table 1). The most recent PacBio protocol, Kinnex[47], the commercial brand of the MAS-seq protocol[25], concatenates cDNA molecules into larger fragment libraries (Fig. 2c), increasing the number of transcripts read in each sequencing pass, thereby increasing throughput without affecting accuracy[25] (Table 1). However, Kinnex/MAS-seq concatenation generates large circularized molecules that must fit into ZMWs. Therefore, it can produce shorter and more truncated transcripts compared with Iso-Seq, the previous standard PacBio library preparation method for RNA sequencing[25,47].

Non-commercial library preparation protocols have been developed to address specific technology issues or biological questions. The ONT-specific R2C2 method creates a circular cDNA sequence that is amplified before ONT sequencing to mimic PacBio's multiple-pass strategy and improve read accuracy[56] (Fig. 2d). The R2C2 method reaches 94–99% read accuracy, making it 1.08–1.14 times more accurate than sequencing the same samples with the same ONT chemistry and basecaller but without R2C2 circularization. However, this comes at the cost of reduced sequencing depth[56,57]. TEQUILA-seq and Capture-seq introduce probe or open reading frame capture to increase throughput of target molecules[58,59] (Fig. 2e). FLAM-seq tails poly(A)-selected RNA with guanosines and inosines for full poly(A) tail capture, before priming for retrotranscription and PCR amplification[60]. Nanopore 3′-end-capture sequencing (Nano3P-seq) uses template switching to initiate RT and capture deadenylated molecules and RNAs with other types of tails[61] (Fig. 2f). To avoid poly(A) annealing biases enriching 5′ degradation products[62], CapTrap-seq modifies the 5′ cap of intact RNA molecules with biotin and uses streptavidin with oligo(dT) priming to detect 5′ capped full-length transcripts[38] (Fig. 2g).

In addition to cDNA, ONT can directly sequence RNA molecules. Although most lrRNA-seq experiments rely on cDNA libraries owing to their high sequencing throughput and accuracy, RT can introduce errors and mis-priming. These issues are often driven by sequence-specific elements in the primary structure of RNA, which can lead to the generation of erroneous cDNA molecules that fail to accurately represent structural variation[63]. By sequencing RNA directly, these issues can potentially be overcome, while enabling the simultaneous detection of epigenetic modifications, which are preserved as the

Table 1 | Data type, length and throughput of long-read sequencing platforms and methods

Technology	Platform	Median read length (kb)	Median throughput per run (10^6 transcript reads)	Refs.
PacBio	cDNA + IsoSeq + Sequel II	~2.1	~2.6	40
	cDNA + Kinnex + Sequel II	~1.7	~40	47
	cDNA + Kinnex + Revio	~1.7	~100	47
ONT	cDNA + MinION (R10.4)	~0.939	~20	40
	cDNA + PromethION (R9.4)	~1	~130	66,67
	dRNA + MinION (R9.4)	~0.8	~1.1	40
	dRNA + PromethION (R9.4)	~0.6	~20	48,232

All values correspond to one single ONT flow cell or one single PacBio SMRTcell. dRNA, direct RNA; ONT, Oxford Nanopore Technologies; PacBio, Pacific Biosciences; SMRT, single-molecule real-time.

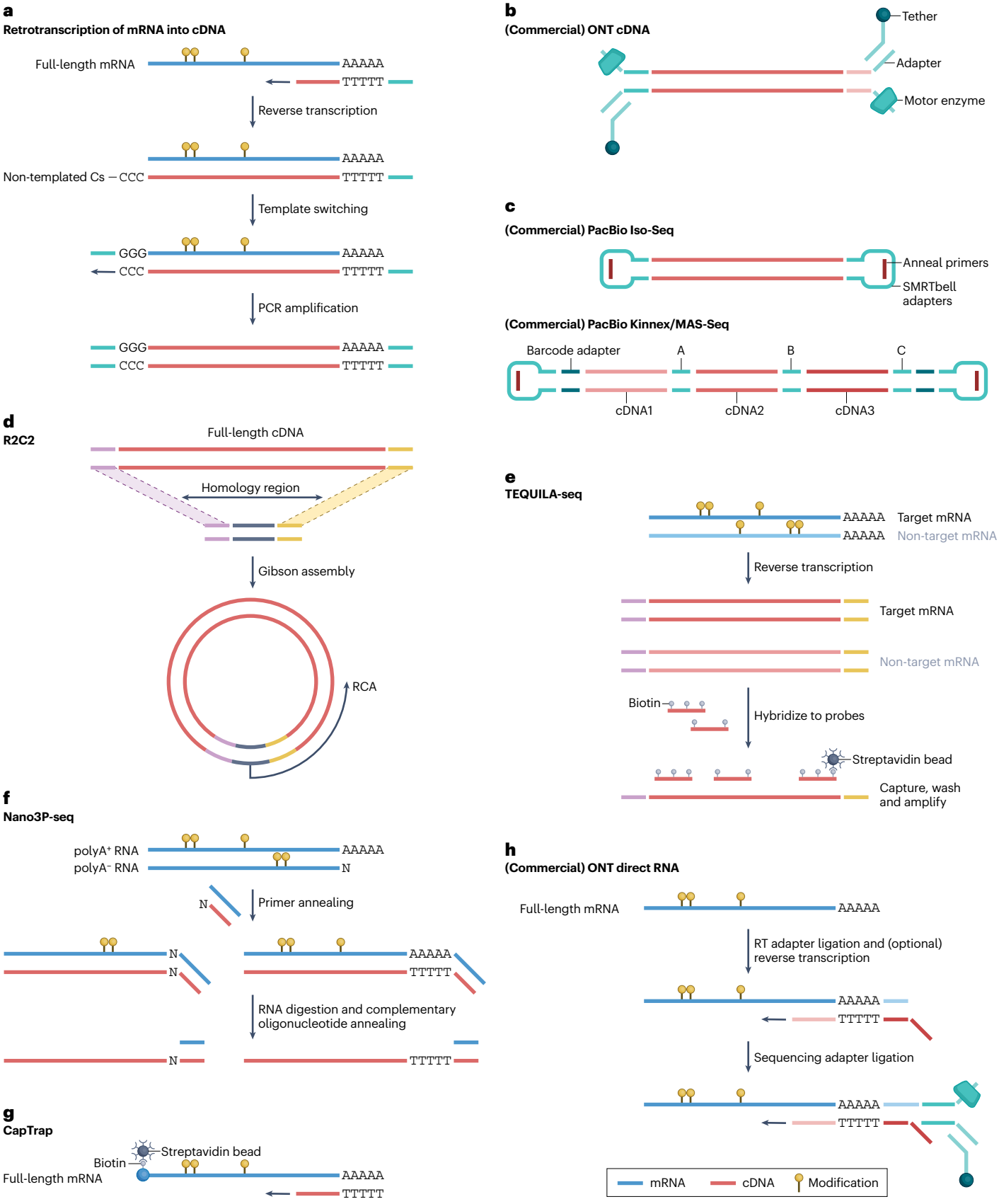

a
Retrotranscription of mRNA into cDNA
Full-length mRNA
AAAAA
TTTTT
Reverse transcription
Non-templated Cs — CCC
Template switching
GGG
CCC
PCR amplification
b
(Commercial) ONT cDNA
Tether
Adapter
Motor enzyme
c
(Commercial) PacBio Iso-Seq
Anneal primers
SMRTbell adapters
(Commercial) PacBio Kinnex/MAS-Seq
Barcode adapter
A
B
C
cDNA1
cDNA2
cDNA3
d
R2C2
Full-length cDNA
Homology region
Gibson assembly
RCA
e
TEQUILA-seq
Target mRNA
Non-target mRNA
Reverse transcription
Biotin
Hybridize to probes
Streptavidin bead
Capture, wash and amplify
f
Nano3P-seq
polyA⁺ RNA
polyA⁻ RNA
N
Primer annealing
RNA digestion and complementary oligonucleotide annealing
g
CapTrap
Biotin
Streptavidin bead
Full-length mRNA
h
(Commercial) ONT direct RNA
RT adapter ligation and (optional) reverse transcription
Sequencing adapter ligation
mRNA
cDNA
Modification

Fig. 2 | Commercial and non-commercial library preparation methods. **a**, Retrotranscription of mRNA into cDNA for sequencing using either Oxford Nanopore Technologies (ONT) or Pacific Biosciences (PacBio). **b**, cDNA library preparation for ONT. **c**, cDNA library preparation for PacBio, including both Iso-Seq and Kinnex/MAS-seq. **d**, R2C2 cDNA library preparation to improve read accuracy in ONT. **e**, TEQUILA-seq library preparation to increase sequencing depth of target molecules. **f**, Nano3P-seq library preparation to capture non-poly(A) tailed RNAs in ONT. **g**, CapTrap-seq library preparation to conserve 5′ ends. **h**, Direct RNA sequencing by ONT for simultaneous detection of transcripts and their modifications. RCA, rolling circle amplification.

RNA does not undergo RT[64]. Direct RNA library preparation involves adding a primer to the 3′ end of RNA molecules and ligating a sequencing adapter[50] (Fig. 2h). Although the sequencing process is simple – the electrical current measured in the nanopore is altered when a modified base passes through – correct assignment of epigenetic modifications is challenging, as it requires machine-learning models pre-trained on ground-truth data[65].

Sequencing depth

The sequencing depth of LRS transcriptomes, measured by the number of reads per sample, has substantially increased in recent years, owing to advances in sequencing platforms, ONT flow cells and PacBio SMRTcells, their chemistries – including the recent RNA004 in ONT and SPRQ in PacBio – as well as more efficient library preparation methods (Table 1). ONT's PromethION and PacBio's Revio instruments can generate up to 130 million and 100 million reads per run, respectively[47,66,67] (Table 1), which has improved the detection and discovery of novel and lowly expressed transcripts[18,25]. The use of adaptive sampling – ONT's method for selectively sequencing transcripts of interest while ejecting non-target transcripts from the pores – has also enabled the enrichment of specific transcripts, improving the detection of known transcripts with low expression levels[68].

The required sequencing depth for an lrRNA-seq experiment depends on the biological question and the diversity of the organism's transcriptome. For example, in *Drosophila*, 2 million versus 250,000 ONT long reads did not have a strong effect on gene detection, but higher sequencing depth increased the discovery of novel transcripts[69]. This finding was also reproduced in a benchmarking experiment studying recall of ground-truth synthetic DNA spike-in controls (sequins) in a full data set versus a down-sampled set[70].

To study known transcripts in human cell lines, brain and heart tissues – in organisms with more complex transcriptomes than *Drosophila* – PacBio recommends generating 10 million high-quality long reads to detect ~80% of known isoforms. Beyond this saturation point, gains in isoform detection diminish[47,71]. However, transcriptomic benchmarking for the recently developed R10.4 ONT flow cells, which differ in read accuracy to PacBio, is pending.

As more transcripts are identified with more reads generated, the detection of rare or unannotated isoforms and accurate transcript quantification require deeper sequencing – >20 million high-quality reads or >40× depth[72,73]. More complex transcriptomes may require even higher depths, with adjusted calculations such as: $\text{Depth}_{\text{sample}} = \frac{\Sigma \text{Depth}_{\text{isoform}}}{N_{\text{expressed isoforms in sample}}}$ (refs. 39,40). Therefore, independent analyses using saturation curves to evaluate organisms with varying transcriptome complexities are necessary to exhaustively evaluate transcriptome coverage and determine optimal library sizes for novel isoform detection, quantification and expression profiling.

Replication and multiplexing

Increased throughput in LRS technologies has enabled the inclusion of biological replicates in lrRNA-seq experiments. Despite recent high correlation (>0.6) across replicates in lrRNA-seq[11,29,40], replicability tends to be lower for transcripts with medium-to-low expression levels. For instance, the Long-read RNA-seq Genome Annotation Assessment Project (LRGASP) benchmark project identified numerous lowly expressed transcripts only in single samples, which were, therefore, not replicated[40]. Replication follows a bimodal distribution, with most transcripts either present in multiple samples or confined to single samples[74], the latter typically novel and lowly expressed RNA molecules. Therefore, replication is necessary for statistical power and to reduce the inclusion of spurious transcripts in downstream analyses.

Previous high costs and low throughput of lrRNA-seq experiments imposed allocating one or more sequencing runs per sample, potentially confounding sequencing run biases with the experimental condition[75–78]. The increased yield of current sequencing platforms facilitates sample multiplexing, and ONT and PacBio provide up to 24 and 12 barcodes to combine samples in a single run (Fig. 1c). Multiplexing thus enables sequencing strategies that distribute experimental conditions across several runs[69]. However, this approach should be carefully implemented, including re-quantifying cDNA after library preparation and adjusting to ensure equimolar concentrations of all samples in the pool.

Ground-truth spike-ins

To assess library preparation and sequencing experiment quality, synthetic spike-in RNA variants (SIRVs)[79], external RNA controls developed by the External RNA Controls Consortium (ERCC)[80] and sequins[81] are used. These preparations vary in RNA length and the presence of alternative isoforms. SIRVs range between ~200 bp and 12 kb and contain alternative transcripts at different concentrations, whereas sequins span from 280 bp to 7,000 bp. Spike-ins provide valuable information about captured transcript length distribution and expected concentrations, but strategies to correct detection and quantification biases have yet to be proposed.

Core data processing of lrRNA-seq data

Core data processing of lrRNA-seq involves raw signal acquisition, basecalling, transcript identification and quantification and quality checks at both long-read and reconstructed-transcript levels (Fig. 1c and Table 2).

Long-read basecalling

Basecalling converts continuous signals – current fluctuations for ONT or light pulses for PacBio – into nucleotide sequences. Advancements in software and hardware have improved accuracy and speed in both technologies. PacBio's DeepConsensus, a six-layer gap-aware transformer, corrects insertion/deletion errors and achieves 99.90% Q30 basecall accuracy, calculated as mean per base accuracy[82], whereas ONT's Dorado achieves 99.26% Q20 basecall accuracy, calculated as median per read accuracy[83]. GPU acceleration yields faster processing times, and both technologies have integrated NVIDIA 100 A GPUs as compatible hardware. Using this equipment, PacBio reduced

Table 2 | Software tools for long-read RNA sequencing data processing and downstream analysis

Tool	Usage	Platform	Refs.
DeepConsensus	Basecalling	PacBio	82
Dorado	Basecalling	ONT	https://nanoporetech.com/platform/accuracy/
Minimap2	Mapping	PacBio, ONT	84
uLTRA	Mapping	PacBio, ONT	85
Graphmap2	Mapping	PacBio, ONT	86
deSALT	Mapping	PacBio, ONT	87
2passtools	Mapping	PacBio, ONT	88
Splam	Mapping	PacBio, ONT	94
Magic-BLAST	Mapping	PacBio, ONT	90
FLAIR	Read clustering Transcript identification and quantification	PacBio, ONT	92
Mandalorion	Read clustering Transcript identification and quantification	PacBio, ONT	31
IsoQuant	Splicing and intron graph construction Alignment correction Transcript identification and quantification Gene-level visualization	PacBio, ONT	32
IsoTools	Splicing and intron graph construction Transcript identification and quantification Differential isoform expression Differential splicing analysis Quality control visualization Gene-level visualization	PacBio, ONT	37
Isosceles	Splicing and intron graph construction Alignment correction Transcript identification and quantification at single-cell, pseudo-bulk and bulk resolution	PacBio, ONT	93
StringTie2	Splicing and intron graph construction Transcript identification and quantification	PacBio, ONT	94
ESPRESSO	Read classification Splice junction correction Transcript identification and quantification Gene-level visualization	PacBio, ONT	95
TALON	Read classification Transcript identification and quantification	PacBio, ONT	96
FLAMES	Read classification Transcript identification and quantification	PacBio, ONT	97
Bambu	Read classification Machine-learning filtering for true novel transcripts Transcript identification and quantification	PacBio, ONT	33
Freddie	Read clustering Transcript identification and quantification	PacBio, ONT	98
LyRic	Transcript identification and quantification	PacBio, ONT	https://github.com/guigolab/LyRic
IsoSplitter	Read clustering Transcript identification and quantification	PacBio, ONT	99
RNA-Bloom2	Splicing and intron graph construction Transcript identification and quantification	PacBio, ONT	100
rnaSPAdes	Splicing and intron graph construction Transcript identification and quantification	PacBio, ONT	101

Table 2 (continued) | Software tools for long-read RNA sequencing data processing and downstream analysis

Tool	Usage	Platform	Refs.
IsoSeq3	Read clustering Read polishing Isoform collapsing Transcript identification and quantification	PacBio	102
RATTLE	Read clustering Read polishing Transcript identification and quantification	ONT	103
isONform	Transcript identification and quantification	PacBio ONT	104
Oarfish	Isoform quantification	PacBio, ONT	105
lr-kallisto	Isoform quantification	PacBio, ONT	34
LIQA	Isoform quantification	PacBio, ONT	35
NanoCount	Isoform quantification	PacBio, ONT	45
LongQC	Read-level quality control Read visualization	PacBio, ONT	106
MinKnow	Read-level quality control Squiggle visualization Read visualization	ONT	https://nanoporetech.com/document/minknow-tech-doc#introduction-to-minknow
SQANTI-reads	Read-level quality control Visualization	PacBio, ONT	69
PycoQC	Read-level quality control	PacBio, ONT	107
NanoPack2	Read-level quality control Mapping-level quality control Read visualization Mapping visualization	PacBio, ONT	108
MultiQC	Read-level quality control Mapping quality control Splice junction quality control	PacBio, ONT	109
SQANTI3	Transcriptome quality control Fusion detection Transcriptome curation and quantification Visualization	PacBio, ONT	46
BUSCO	Genome completeness quality control	PacBio, ONT	110
IsoSeqSim	PacBio, ONT	PacBio, ONT	111
Trans-NanoSim	Long-read simulator	ONT	112
SQANTI-SIM	Novel transcript simulator	PacBio, ONT	113
DESeq2	Differential gene expression Differential isoform expression	Short-read RNA sequencing (srRNA-seq), PacBio, ONT	114
edgeR	Differential gene expression Differential isoform expression	srRNA-seq, PacBio, ONT	115
limma-voom	Differential gene expression Differential isoform expression	srRNA-seq, PacBio, ONT	116,117
DELongSeq	Differential isoform expression	PacBio, ONT	36
TAGET	Differential isoform expression	PacBio, ONT	118
tappAS	Differential isoform expression Differential splicing analysis Functional profiling Gene-level visualization	PacBio, ONT	119

Table 2 (continued) | Software tools for long-read RNA sequencing data processing and downstream analysis

Tool	Usage	Platform	Refs.
HBA-DEALS	Differential isoform expression Differential splicing analysis	PacBio, ONT	120
ScisorseqR	Differential isoform expression Differential splicing analysis	PacBio, ONT	10
DEXSeq	Differential transcript usage	srRNA-seq, PacBio, ONT	121
Augustus	Gene prediction Genome annotation	PacBio, ONT	134,135
SNAP	Gene prediction Genome annotation	PacBio, ONT	136
MAKER	Gene prediction Genome annotation	PacBio, ONT	137
BRAKER	Gene prediction Genome annotation	PacBio, ONT	138
LongGF	Gene fusion discovery	PacBio, ONT	143
JAFFAL	Gene fusion discovery	PacBio, ONT	144
Genion	Gene fusion discovery	PacBio, ONT	145
FusionSeeker	Gene fusion discovery	PacBio, ONT	12
FUGAREC	Gene fusion discovery	PacBio, ONT	147
CTAT-LR-fusion	Gene fusion discovery	PacBio, ONT	147
FLAIR-fusion	Gene fusion discovery	PacBio, ONT	148
APPRIS	Isoform functional annotation	PacBio, ONT	157
FunctionAnnotator	Isoform functional annotation	PacBio, ONT	157
ISOGO	Isoform functional annotation	PacBio, ONT	159
IsoAnnot	Isoform functional annotation	PacBio, ONT	119
IsoAnnotLite	Isoform functional annotation	PacBio, ONT	46
ExonOntology	Exon skipping detection Functional profiling	PacBio, ONT	161
DIGGER	Exon skipping detection Functional profiling	PacBio, ONT	162
IRFinder-S	Intron retention detection Functional profiling	PacBio, ONT	163
IsoformSwitchAnalyzeR	Differential splicing analysis Functional profiling	PacBio, ONT	164
DoChaP	Differential splicing analysis Functional profiling	PacBio, ONT	165
NEASE	Differential splicing analysis Functional profiling	PacBio, ONT	166
LORALS	Allele-specific quantification	PacBio, ONT	18
RPVG	Allele-specific quantification	PacBio, ONT	172
IsoLaser	Allele-specific quantification	PacBio, ONT	173
FLAIR2	Transcript identification Allele-specific quantification	PacBio, ONT	20
UCSC genome browser	Read visualization Mapping visualization	PacBio, ONT	174
Integrative Genomics Viewer	Read visualization Mapping visualization	PacBio, ONT	175
SMRT Link	Read-level quality control Read visualization	PacBio	https://www.pacb.com/smrt-link/

Table 2 (continued) | Software tools for long-read RNA sequencing data processing and downstream analysis

Tool	Usage	Platform	Refs.
BulkVis	Squiggle visualization	ONT	176
SquiggleKit	Squiggle visualization	ONT	177
BoardION	Squiggle visualization	ONT	178
Swan	Gene-level visualization	PacBio, ONT	179
ggtranscript	Gene-level visualization	PacBio, ONT	179
ScisorWiz	Gene-level differential isoform expression visualization	PacBio, ONT	181
R2Dtool	Gene-level visualization	PacBio, ONT	182
Nanoblot	Gene-level visualization	PacBio, ONT	183
modbamtools	RNA modification visualization	ONT	184
Methylartist	RNA modification visualization	ONT	185
NanoMethViz	RNA modification visualization	ONT	186
NanoTrans	Differential expression visualization RNA modification visualization Gene-level visualization	ONT	187
ELIGOS	RNA modification detection	ONT	22
xPore	RNA modification detection	ONT	189
DRUMMER	RNA modification detection	ONT	190
nanom6A	RNA modification detection	ONT	21
m6Anet	RNA modification detection	ONT	191
DENA	RNA modification detection	ONT	192
Penguin	RNA modification detection	ONT	193
mAFiA	RNA modification detection	ONT	194
NanoSpa	RNA modification detection	ONT	195
TandemMod	RNA modification detection	ONT	196
Xron	RNA modification detection	ONT	197
m6ABasecaller	RNA modification detection	ONT	198
Nanopolish	Segmentation	ONT	199
Tombo	Segmentation	ONT	200
f5c	Segmentation	ONT	201
Uncalled4	Segmentation	ONT	202
MasterOfPores	RNA modification detection pipeline	ONT	204
nf-core/nanoseq	RNA modification detection pipeline	ONT	https://nf-co.re/nanoseq/3.1.0/
tailfindr	Poly(A) detection via segmentation	ONT	207

ONT, Oxford Nanopore Technologies; PacBio, Pacific Biosciences.

circular consensus sequencing times from 15 h to 2.5 h, according to NVIDIA. Further improvements in accuracy and speed are expected as technologies evolve.

Mapping long reads to reference genomes

Mapping long reads or transcripts to the reference genome or transcriptome presents challenges owing to the presence of splice sites and sequencing errors that require mappers capable of handling split alignments. Minimap2, the recommended and most widely used mapper for long and noisy sequences, uses an efficient seed-and-chain framework[84] to extract *k*-mers from the reference genome and index them to identify exact matches (anchors) with query sequences. However, Minimap2 has limitations when aligning small exons and handling complex split alignments across variable splice junctions[85].

To address these specific issues in studies focused on small exons or complex split alignments, specialized aligners have been developed: uLTRA improves splice alignment accuracy, especially for small exons and complex junctions, by using a two-pass collinear chaining algorithm that leverages exon annotations for precise boundary detection[85]; Graphmap2 optimizes anchor selection and refines alignments at exon boundaries[86]; deSALT uses a de Bruijn graph-based approach with a two-pass alignment to generate spliced reference sequences and produce refined alignments[87]; 2passtools and Splam incorporate machine learning to filter spurious splice junctions[88,89];

and Magic-BLAST chains collinear local alignments to maximize scores and maintain splice signal consistency, mapping reads to genome and transcriptome without a two-pass process[90].

Transcript identification and quantification

Transcript identification and quantification are the primary goals of lrRNA-seq, which can be achieved through two main strategies: inferring transcript models from the long reads, with or without the support of a reference, and quantifying them based on the model; or quantifying transcripts based on existing genome annotations by mapping or pseudo-aligning reads to a reference transcriptome[34]. Transcript model quantification enables novel transcript discovery but is usually computationally intensive, whereas quantification based on a reference transcriptome by definition does not include novel transcript identification but is computationally efficient. The approach to choose depends on the goal of the study (Box 1), but it should be noted that lrRNA-seq is characterized by the detection of a large number of novel transcripts (albeit of low expression)[40], implying that reference annotations are expected to be, in general, incomplete, which can affect read assignment to reference transcripts[91].

Algorithms for transcript identification followed by quantification can be classified into cluster-based, graph-based and class-based methods (Fig. 3a), further distinguished by their dependence on a reference genome or annotation into reference annotation-free and fully reference-free methods. Cluster-based methods such as FLAIR[92] and Mandalorion[31] align reads to a reference genome, grouping them based on shared splice junctions or chains. Graph-based approaches, such as IsoQuant[32], IsoTools[37], Isosceles[93] and StringTie2[94], construct splice graphs that represent the splicing landscape of the transcriptome. IsoQuant emphasizes alignment correction and de novo transcript discovery; IsoTools focuses on alternative splicing analysis; Isosceles is tailored for single-cell isoform discovery; and StringTie2 can use short reads to handle errors. Class-based methods, such as ESPRESSO[95], TALON[96], FLAMES[97] and Bambu[33], classify reads by splice junction patterns to identify known, novel or uncertain isoforms. ESPRESSO corrects splice junctions before classification; TALON validates novel transcripts across replicates; FLAMES analyses incongruent reads for novel splice sites; and Bambu uses machine learning to predict and rank novel transcripts while controlling false positives.

Annotation-free methods, such as Freddie[98] and LyRic, detect and quantify transcript isoforms without existing annotations (Fig. 3a). Freddie optimizes clustering and error correction for isoform reconstruction, whereas LyRic returns fewer but high-confidence transcript models supported by short reads[40]. Fully reference-free methods are

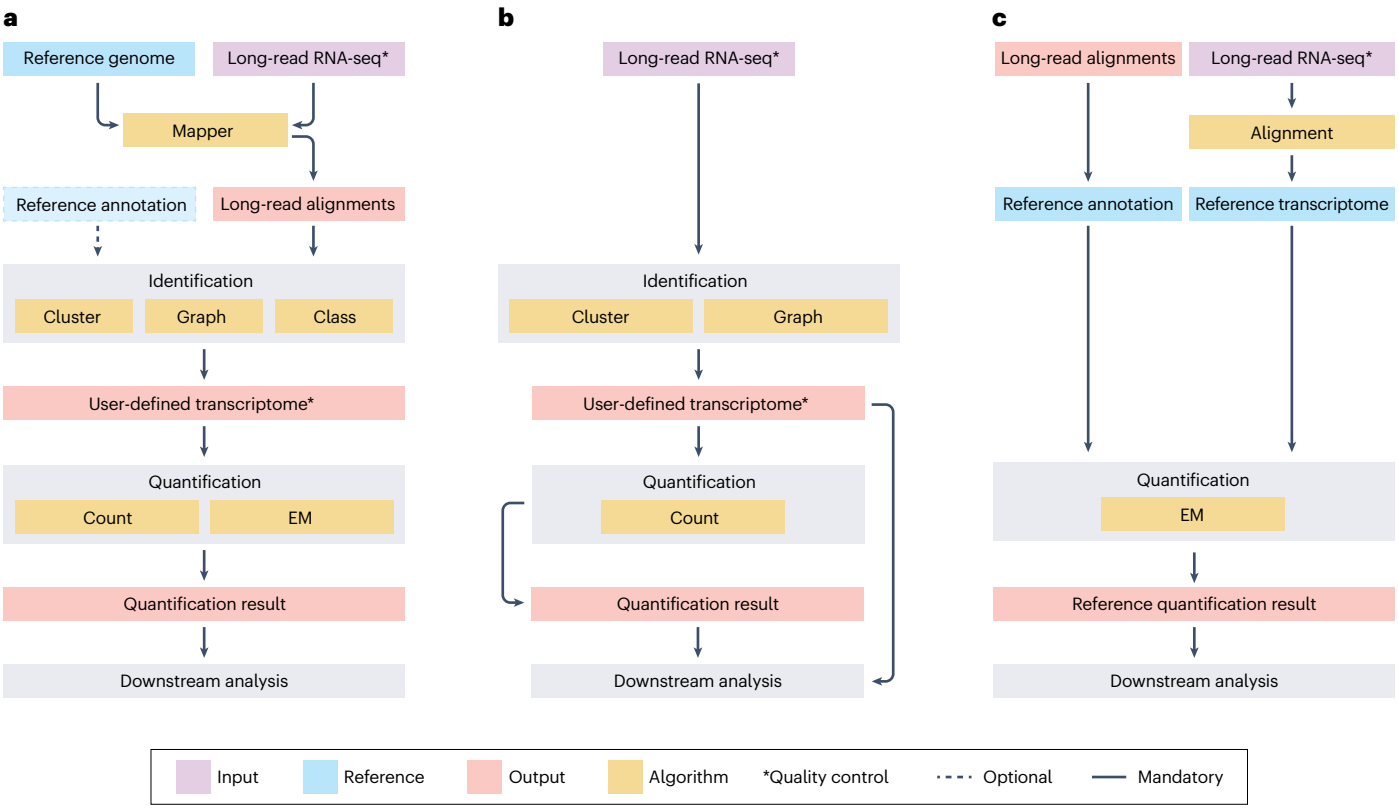


Fig. 3 | Approaches for transcript identification and quantification using long-read RNA sequencing data. a, Reference-based inference. This method creates an experiment-defined transcriptome by mapping reads to a reference genome. Transcript identification can apply cluster-based, graph-based or class-based algorithms and may or may not use an existing reference annotation to guide analysis. The approach allows the discovery of novel transcripts and results in a long-read-defined transcriptome that is quantified for secondary analysis. **b**, Reference-free inference. This approach entirely relies on long-read RNA sequencing data for transcript reconstruction and quantification and is suitable for species without reference information. A mapping step may be used to join transcripts as alternative isoforms if a genome sequence is available. **c**, Annotation-driven. In this case, the reference annotation or transcriptome is used for direct transcript quantification by (pseudo)mapping, resulting in a simpler and faster process but missing any novel isoforms. EM, expectation-maximization algorithm for isoform quantification.

valuable when both reference genomes and annotations are unavailable (Fig. 3b), using clustering techniques to group similar reads, detect alternative splicing events and refine transcript models: IsoSplitter validates splice junctions with short reads[99]; RNA-Bloom2 (ref. 100) and rnaSPAdes[101] use de Bruijn graphs for assembly; IsoSeq3 (ref. 102) refines PacBio HiFi reads to call transcripts that are then mapped to the genome to identify alternative isoforms of the same gene; RATTLE uses *k*-mer-based error correction for ONT data[103]; whereas isONform uses a directed acyclic graph to preserve actual splicing variations[104].

Direct quantification tools face challenges in uniquely assigning reads to specific isoforms owing to sequencing errors, length variability and RNA transcript complexity. To address these limitations, they rely on reference annotations and the expectation-maximization algorithm, an approach to perform maximum likelihood estimation with latent variables, to iteratively refine isoform assignments based on abundances (Fig. 3c). For example, Oarfish uses coverage-based probabilistic modelling to adjust fragment assignment probabilities, addressing uneven coverage in long-read data sets[105]. Similarly, lr-kallisto adapts the pseudo-alignment strategy to accommodate longer reads and higher error rates, focusing on reducing the complexity of transcript compatibility graphs for speed and resource efficiency[34]. LIQA introduces a bias correction model to handle truncated reads and 3′-end biases by fitting them into a Kaplan–Meier estimator, enhancing isoform quantification accuracy[35]. NanoCount uses stringent alignment scores and 3′-end position filtering to improve read assignment reliability for ONT direct RNA sequencing[45].

Quality control

Despite improvements in ONT and PacBio sequencing platforms, lrRNA-seq data quality is still influenced by library preparation, instrument performance and data processing choices. Comprehensive quality control (QC) is essential to assess potential biases in the sequencing output and reliability of the inferred transcript models.

At the read level, tools such as LongQC[106], MinKnow, SQANTI-reads[69] and PycoQC[107] obtain QC metrics including read counts, length distribution, basecalling quality scores, adapter contamination, GC content and coverage. Furthermore, at the read alignment level, NanoPack2 offers modules for read-level QC and assessing the quality of mapped reads to the reference genome[108]. MultiQC provides an overview of these read and alignment-level metrics[109], whereas SQANTI-reads integrates assessments of transcriptome QC across multiple samples to detect outliers and estimate discovery power using the SQANTI3 framework. SQANTI3 is a widely adopted approach to QC LRS-based reconstructed transcriptomes, as it defines transcript structural categories of novelty and integrates complementary data – short reads, cap analysis of gene expression peaks, poly(A) motifs and poly(A) sites – to assess confidence and curate transcripts[46].

If synthetic spike-ins have been included in the library preparation, they can be used to evaluate data quality and quantification in lrRNA-seq. ERCC controls mimic mono-exonic RNAs of varying lengths and concentrations, facilitating assessment of RNA length biases and gene-level quantification accuracy. Additionally, SIRVs and sequins mimic alternatively spliced transcripts and are helpful for evaluating the identification of splice sites, alternative splicing and transcript-level quantification. Furthermore, SIRVs are provided in different mixes with varying relative concentrations, offering a reference for quantification assessment[40].

To perform QC on the efficiency of the transcriptome capture in the experiment, bioinformatics benchmarking tools such as BUSCO compare a genome annotation file against a set of conserved single-copy orthologues, providing a quantitative measure of completeness[110]. However, BUSCO was initially designed to evaluate genome assemblies and focused on single-copy genes, limiting its ability to assess alternative isoforms, which are often reported as duplicated genes.

Finally, to evaluate whether sample-specific data processing and analysis strategy are adequate, algorithms that simulate ONT and PacBio lrRNA-seq data[111,112] or novel transcripts[113] are valuable. They facilitate the benchmarking of tools used for alignment, quantification and sensitivity to transcript identification, among other metrics. However, they may not accurately reproduce all possible errors, such as RNA degradation, library preparation artefacts and limited capture. Therefore, new methods are needed for comprehensive ground-truth assessment of lrRNA-seq that account for biases and transcriptome complexity.

Downstream computational analysis

The downstream analysis of lrRNA-seq data includes approaches to obtain biological insights from the data. The most frequent applications are differential expression analysis, genome annotation and RNA modification discovery (Fig. 1c and Table 2).

Differential isoform expression and splicing

Recent increases in sequencing throughput enable lrRNA-seq for differential gene and isoform expression analyses and uniquely position LRS technologies for differential isoform usage studies (Fig. 4a). A recent benchmark[70] evaluated the suitability of srRNA-seq tools using synthetic lrRNA-seq data sets. Although all tools effectively controlled for false discoveries, DESeq2 (ref. 114), edgeR[115] and limma-voom[116,117] outperformed other tools for differential isoform expression using LRS. However, LRS presents unique challenges, including transcript selection biases, sequencing error rates and mapping accuracy. To address the challenges of lrRNA-seq data, new tools have been proposed. DELongSeq detects differential isoform expression based on expression and uncertainty estimates instead of read counts using a random-effect regression model[36]. TAGET uses an exact two-sided binomial test on the observed transcript counts to conduct differential isoform expression analysis[118].

Differential splicing analysis can detect changes in splicing patterns that contribute to transcript diversity and potentially affect gene function (Fig. 4a). Tools such as IsoTools, tappAS, HBA-DEALS and ScisorseqR can perform both differential isoform expression and splicing analysis[37,119,120]. IsoTools uses the method by Joglekar et al.[29], applying a χ^2 test on an isoform quantification dataframe to identify genes with differential isoform expression between samples, and a likelihood ratio test for differential splicing[37]. tappAS uses DEXseq[121] for differential splicing and introduced the 'total isoform usage change' as a measure of the magnitude of redistribution of isoform expression between conditions, comparing it with gene fold-change to evaluate the relative contribution of gene expression and splicing transcriptional regulation. HBA-DEALS models differential expression and splicing simultaneously by using hierarchical Bayesian analysis[120], whereas ScisorseqR analyses differential poly(A), TSS and exon usage using a χ^2 test on a feature quantification dataframe[10] (Table 2).

Despite great advances in testing the suitability of short-read methods and new differential isoform expression and splicing analysis methods for lrRNA-seq, technical and biological biases that affect transcript quantification can markedly influence analysis outcomes.

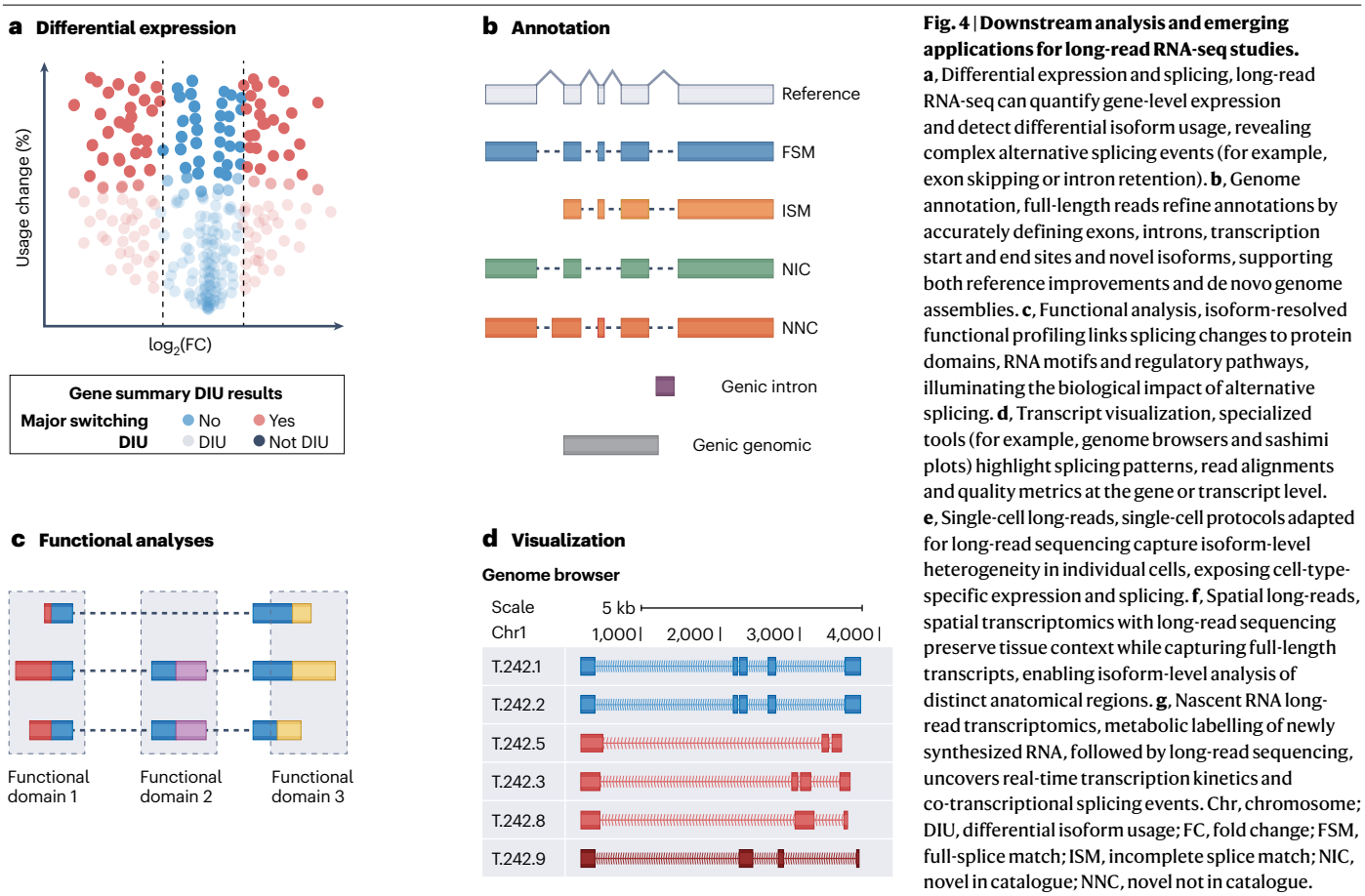


Fig. 4 | Downstream analysis and emerging applications for long-read RNA-seq studies. **a**, Differential expression and splicing, long-read RNA-seq can quantify gene-level expression and detect differential isoform usage, revealing complex alternative splicing events (for example, exon skipping or intron retention). **b**, Genome annotation, full-length reads refine annotations by accurately defining exons, introns, transcription start and end sites and novel isoforms, supporting both reference improvements and de novo genome assemblies. **c**, Functional analysis, isoform-resolved functional profiling links splicing changes to protein domains, RNA motifs and regulatory pathways, illuminating the biological impact of alternative splicing. **d**, Transcript visualization, specialized tools (for example, genome browsers and sashimi plots) highlight splicing patterns, read alignments and quality metrics at the gene or transcript level. **e**, Single-cell long-reads, single-cell protocols adapted for long-read sequencing capture isoform-level heterogeneity in individual cells, exposing cell-type-specific expression and splicing. **f**, Spatial long-reads, spatial transcriptomics with long-read sequencing preserve tissue context while capturing full-length transcripts, enabling isoform-level analysis of distinct anatomical regions. **g**, Nascent RNA long-read transcriptomics, metabolic labelling of newly synthesized RNA, followed by long-read sequencing, uncovers real-time transcription kinetics and co-transcriptional splicing events. Chr, chromosome; DIU, differential isoform usage; FC, fold change; FSM, full-splice match; ISM, incomplete splice match; NIC, novel in catalogue; NNC, novel not in catalogue.

Until these ambiguities are resolved, we recommend performing the quantification and differential expression/splicing steps using more than one tool and using the intersection of their results.

Genome annotation

lrRNA-seq provides full-length information of gene expression and is therefore an interesting source of data for genome annotation efforts. SQANTI3 provides a primary annotation framework by annotating transcripts upon comparison to reference annotations into full-splice-match (that is, reference transcripts), incomplete-splice-match (that is, fragments and alternative TSS/TTS), novel-in-catalogue (that is, alternative transcripts of current annotations), novel-not-in-catalogue (that is, transcripts with novel splice sites) and others[46] (Fig. 4b). lrRNA-seq is currently used by the two main reference human gene and transcript annotation resources, NCBI's RefSeq[122,123] and EMBL-EBI's Ensembl/GENCODE[124,125], to complete reference genome annotations, including condition-specific rare transcripts[74]. Also, it is used by multiple international initiatives that aim to sequence all species on Earth (for example, Earth Biogenome[126], the Vertebrate Genome Project[127] and the Darwin Tree of Life[128]). Genome annotation approaches using lrRNA-seq data are evidence-based or evidence-driven.

Evidence-based methods use experimental data such as RNA-seq or protein sequences to identify genes by mapping them to the genome. lrRNA-seq captures full-length transcripts, making it the preferred choice for high-confidence genome annotation. This method improved annotations for model organisms, such as *Arabidopsis thaliana*[19], and less well-characterized species, such as the horse (*Equus ferus caballus*)[129], the tea plant (*Camellia sinensis*)[130] and the Indian jumping ant (*Harpegnathos saltator*)[131].

Evidence-driven strategies combine computational models such as support vector machines or hidden Markov models to predict genes based on genomic sequences[132,133] and experimental data. Tools including Augustus[134,135], SNAP[136], MAKER[137] and BRAKER[138] can leverage lrRNA-seq data for genome annotation. A recent strategy used to annotate the Florida manatee (*Trichechus manatus latirostris*) genome summarized reads into transcript models ahead of using evidence-driven methods, improving annotation by reducing gene redundancy and accurately defining boundaries of TSS, exons and TTS[139].

Gene fusion discovery

Gene fusions, which result from *cis*-splicing or *trans*-splicing or genomic rearrangements, are important cancer drivers and diagnostic markers[140–142]. lrRNA-seq detects transcripts that span two genes and enables sequencing of full-length fusion transcripts. Gene fusion discovery tools (Table 2) use splice-aware aligners to map reads to reference genomes or transcriptomes and identify reads spanning multiple genes. SQANTI3 identifies fusion transcripts as those mapping to two genes after Minimap2 alignment[46]. LongGF[143] and JAFFAL[144] determine fusion breakpoints from these alignments and assign a confidence score based on the number of supporting reads. Genion[145] and FusionSeeker[12] refine fusion detection by clustering reads that suggest potential fusions, and FUGAREC[146] and CTAT-LR-fusion[147] validate fusion positions by re-aligning reads to synthetic gap sequences or collinear gene contigs. Finally, FLAIR-fusion[148] is able to extract isoform-level fusions by annotating the transcriptome after performing the genomic alignment and refining the isoforms using FLAIR[92]. In general, the ability of lrRNA-seq to sequence full-length transcripts greatly increases the confidence in gene fusion discovery.

Functional annotation

Functional annotation assigns biological meaning to sequences to understand the impact of gene expression regulation. It involves identifying protein-coding transcripts, predicting functional domains and annotating them using orthologue information from databases such as SwissProt[149], Pfam[150] and InterPro[151]. Gene ontology terms and biological pathways can also be mapped[152–154]. Databases such as Rfam[155] and MirBase[156] are used to annotate RNA families. These resources support similarity-based annotation; however, such gene-centric approaches do not account for the functional consequences of differential splicing.

Databases such as APPRIS[157], FunctionAnnotator[158] and ISOGO[159] provide functional annotations with isoform resolution. However, these annotations are often based on protein-level features, overlooking post-transcriptional regulatory mechanisms. To address these limitations, IsoAnnot[119] and IsoAnnotLite[46] use long-read-defined transcriptomes to build an isoform-resolved database of functional annotations at both RNA and protein levels, enabling the annotation of novel isoforms. Despite advances in functional annotation, profiling non-coding transcripts such as long non-coding RNAs remains challenging owing to their lower conservation across species.

Functional profiling

Transcriptomic studies typically aim to characterize the molecular functions and pathways of differentially expressed genes (Fig. 4c). srRNA-seq achieves this goal by identifying over-represented functions among differentially expressed genes[160]. However, alternative splicing has a substantial impact on tissue identity and cell development. To investigate the functional consequences of alternative splicing, exon skipping events can be detected using Exon Ontology or DIGGER to assess their impact on protein structure and function[161,162], and IRFinder-S leverages lrRNA-seq to identify intron retention events[163]. Other tools focus on transcript-level analysis. For example, IsoformSwitchAnalyzeR statistically tests for changes in genome-wide alternative splicing patterns and consequences of isoform switches[164], whereas tappAS uses isoform-resolved annotation of coding and non-coding functional domains, motifs and sites to evaluate the functional impact of transcript variants and post-transcriptional regulation on isoform expression. Moreover, the tool introduces 'differential feature inclusion analysis', which assesses the inclusion or exclusion of a wide range of functional motifs resulting from alternative splicing[119]. DoChaP and NEASE use exon-protein domain associations and conservation to explore the structural effects of alternative splicing in protein domains[165,166]. Regardless of the approach, obtaining accurate functional profiling critically depends on having well-annotated isoforms.

Allele-specific expression

Differences in the expression of allelic variants can contribute to phenotypic variation[18,167]. For many years, linear or single reference genomes complicated the study of how common and rare genetic variants influence regulatory effects on the transcriptome, introducing mapping biases because reads could not be accurately aligned to the specific haplotypes from which they originated. However, with the growing capabilities of LRS, efforts have intensified to develop pan-genome and pan-transcriptome references[74,168–170].

By capturing full-length transcripts, lrRNA-seq improves the mapping of heterozygous genetic variants and enables the detection of allele-specific expression to analyse allelic imbalance and alternative splicing[171] (Fig. 1b). LORALS and RPVG use pan-genomes as mapping

references to phase haplotypes[18,172]. Although LORALS statistically tests whether the observed expression is higher than expected noise levels[18], RPVG infers the most probable underlying haplotype pairs and estimates allele-specific expression using an expectation-maximization approach[172]. IsoLaser and FLAIR2 analyse allele-specific alternative splicing by generating phased haplotypes from variant calling within lrRNA-seq reads[20,173]. IsoLaser tests for linkage between the phased haplotypes and alternatively spliced exon segments[173], whereas FLAIR2 aims to detect haplotype-specific isoforms and generates transcript models with accurate TSS and TTS[20]. Together, these methods exemplify how leveraging haplotype-aware references and lrRNA-seq can uncover allele-specific expression and splicing dynamics.

Visualization

Visualization of lrRNA-seq data is commonly done using the UCSC genome browser[174] or the Integrative Genomics Viewer (IGV)[175] (Fig. 4d). However, numerous specialized tools have been designed to visualize different elements of interest throughout the data processing pipeline (Table 2). ONT and PacBio's proprietary software, MinKNOW and SMRT Link, respectively, provide real-time run monitoring and summary plots of read quality. Other tools include BulkVis, SquiggleKit and BoardION to plot electric current squiggles of ONT runs[176–178]; LongQC to plot quality metrics of FASTQ files[106]; and NanoPack2 for both FASTQ and BAM files[108]. SQANTI3 offers plots to inspect QC features of the data, for example, the distance to TSS or TTS, incidence of non-canonical splicing or intrapriming levels (an artefact arising from hybridization of oligo(dT) primer onto adenine single-nucleotide repeats, in addition to poly(A) tails, which can overinflate mRNA counts for genes containing these repeats)[46], whereas SQANTI-reads provides comparative charts of quality features across multiple samples[69].

Other tools focus on gene-level visualizations. Swan[179] creates gene and transcript path graphs, whereas ggtranscript highlights differences from the reference transcript[180]. ScisorWiz visualizes gene-level differential isoform expression[181], and R2Dtool and Nanoblot highlight isoform-resolved RNA features[182,183]. Moreover, IsoTools[37], ESPRESSO[95], IsoQuant[32] and tappAS[119] provide a range of plots, from data QC, including saturation plots and transcript quality, to gene-level splicing visualizations using sashimi plots. Finally, specialized packages for analysing ONT direct RNA generate plots related to differential isoform expression, RNA epigenetic modifications[184–186], poly(A) tails and gene fusions[187] (Table 2). As the transcriptomics community continues to grow, new tools with enhanced features are expected to emerge.

RNA modification detection

Chemical modifications of RNA, collectively known as the epitranscriptome, have crucial roles in various regulatory processes (reviewed elsewhere[188]). ONT direct RNA sequencing enables the detection of unique signatures of these RNA modifications in individual RNA molecules[50] (Fig. 1b). Algorithms for detecting epitranscriptomic modifications via ONT can be classified into three groups: unsupervised, supervised or semisupervised methods. Unsupervised or comparative methods, such as ELIGOS[22], xPore[189] and DRUMMER[190], analyse basecalled data to identify modifications by statistically testing whether the signal differs between modified and unmodified samples; whereas ELIGOS and DRUMMER rely on basecalling error signals and xPore directly assesses the ONT ion current signal. Supervised methods, including nanom6A[21], m6Anet[191], DENA[192], Penguin[193] and mAFiA[194], use machine-learning models trained on ground-truth, in vitro-generated or labelled data sets containing known RNA modifications. Some supervised tools, such as NanoSpa[195] and TandemMod[196], can detect multiple types of RNA modifications within the same molecule. Semi-supervised methods such as Xron[197], m6ABasecaller[198] and Dorado are trained on both immunoprecipitation and in vitro transcription data to enable de novo identification of m^{6}A modifications in single reads with single-nucleotide resolution at the basecalling step. In many cases, RNA modification detection methods require a data pre-processing step called 'segmentation' where the raw ONT ion current signal is aligned to a reference to accurately define event boundaries. Segmentation can be performed using tools such as Nanopolish[199], Tombo[200], f5c[201] or Uncalled4 (ref. 202). Despite the advances made by the three mentioned types of RNA modification callers, many signal modulations have not yet been associated with lesser-known RNA modifications, owing to technical biases[51] and the challenge of generating synthetic data sets that capture broader modification diversity[203].

Although direct RNA sequencing has expanded our ability to study the epitranscriptome, few comprehensive, end-to-end pipelines are available, with the exception of MasterOfPores[204] and nf-core/nanoseq. Although some benchmarking studies have been conducted to compare m^{6}A profiling methods[205], a comprehensive assessment of tools for single and simultaneous detection of other chemical RNA modifications using robust ground-truth data sets remains a marked unmet need.

In addition to epitranscriptomic modifications, other post-transcriptional modifications, such as transcript poly(A) tails, can be analysed using lrRNA-seq[60,64]. Although shorter poly(A) tails are thought to be associated with more stable mRNA, longer poly(A) tails are associated with increased translational efficiency[206]. Poly(A) tails can be directly extracted from basecalled PacBio cDNA sequencing data[60] and can also be inferred from direct RNA sequencing with ONT[64]. However, in ONT experiments, basecallers struggle to accurately call homopolymeric regions. Thus, specialized tools that take advantage of a segmentation step by comparing the raw ONT ion current signal with a poly(A) reference, such as Nanopolish-polyA[199] and the more recent tailfindr[207], have been developed and are widely used to measure poly(A) tail lengths.

Emerging lrRNA-seq applications

lrRNA-seq has rapidly expanded its range of applications with the development of protocols that address transcript expression at diverse resolutions and cellular compartments. Here, we discuss long-read single-cell methods and LRS-based transcription and translation analyses (Figs. 1c and 4e–g). These examples illustrate the power of LRS to be combined with various biochemical assays to investigate the biogenesis, processing and metabolism of RNA with isoform resolution.

Long-read single-cell RNA-seq

Long-read single-cell RNA-seq (LR-scRNA-seq), for example, by 10× Genomics, leverages droplet-based methods with cell barcodes and unique molecular identifiers (UMIs)[208] (Fig. 4e). In brief, the strategy is to direct 10× cDNA library preparations to LRS, rather than to srRNA-seq, to obtain reads of full-length transcripts (reviewed elsewhere[209]). However, high error rates in ONT and low sequencing depth in PacBio can lead to inaccurate barcode and UMI assignments and limited low-abundance transcript detection. Sicelore and RAGE-seq combine LR-scRNA-seq with srRNA-seq to improve barcode and UMI assignment accuracy but increase time and cost[210,211]. scTaILoR-seq enriches genes via targeted hybridization capture[212], and computational tools such as scNanoGPS[213] and BLAZE[214] improve

read assignment and barcode verification. The PacBio's Kinnex kit and Revio system boost throughput to 80–100 million reads per SMRT cell[47] (Table 1), whereas HIT-scISOseq reduces sequencing artefacts using biotinylated primers and magnetic beads[215]. LR-scRNA-seq has revealed complex isoform dynamics in different tissues, including brain-specific differential isoform expression events in 395 genes across 45 cell types between the mouse hippocampus and prefrontal cortex[10].

Long-read spatial transcriptomics

Long-read spatial transcriptomics uses methods such as 10× Genomics' Visium, Slide-seq[216] and Stereo-seq[217] for in situ spatial barcode capture and integration with LRS platforms[208] (Fig. 4f), with similar challenges and solutions as proposed for LR-scRNA-seq[218]. Further developments in LRS-based spatial transcriptomics focus on enhancing in situ capture by balancing the trade-off between spatial resolution and field of view and advancing sequencing depth and accuracy to enable srRNA-seq-free barcode and UMI assignment[30]. Interestingly, the technology has identified region-specific isoform variations in key genes[219] and transcriptional heterogeneity within glioma microenvironments, revealing stem-like cells in invasive glioma niches[218].

Nascent lrRNA-seq

Nascent RNA sequencing offers insights into gene regulation, RNA polymerase II (Pol II) dynamics and transcription-coupled splicing (reviewed elsewhere[220]) (Fig. 4g). Chromatin-associated RNA isolation uses stable nascent RNA–Pol II interactions, sequencing both nascent and stable RNAs, but lacks Pol II immunoprecipitation and relies on RT and PCR, hindering RNA modification detection and possibly affecting 5′ ends[221–224]. POINT-nano uses Pol II-associated immunoprecipitation for specific nascent RNA enrichment[225], whereas dNET-seq, Nano-COP isolate new RNA via biotinylation[226,227], and Nano-ID uses neural networks for nascent RNA identification[228]. These techniques revealed that splicing occurs near active Pol II in yeast, within tens of nucleotides downstream of the 3′ splice site[222,224]. In mammals, this approach showed that splicing can happen as early as 15 nt downstream of Pol II, extending up to 300 nt (ref. 225), revealing that splicing can be completed before Pol II reaches the next splice site or delayed until more intron is transcribed[225–227]. Furthermore, LRS enhances understanding of splicing kinetics, showing that faster transcription favours exon skipping, whereas slower Pol II activity favours exon inclusion, which aligns with the kinetic competition model[224]. Precise Pol II mapping suggests that splicing decisions occur after transcribing the downstream exon, supporting the exon definition model[224].

Long-read ribosomal sequencing

Translational profiling bridges mRNA transcription and protein synthesis, and recent LRS-based methods have been proposed to study isoform-level translation. Specifically, the long-read Ribo-STAMP (lr-Ribo-STAMP) method transfects cells with a fusion protein construct consisting of ribosomal protein S2 (RPS2) fused to APOBEC1, an RNA editing enzyme that catalyses cytidine (C) to uridine (U) editing on translating transcripts. The subsequent LRS and identification of the edited transcript enable simultaneous measurements of mRNA translation and mRNA levels[229]. This approach was applied to triple-negative breast cancer cell line under normal oxygen levels and hypoxia to find transcriptional and translation differences between the two conditions as well as regulatory elements that mediate translational differences at the isoform level[229].

Conclusions and future perspectives

lrRNA-seq is quickly becoming the preferred method for transcriptome analysis, with improved sequencing accuracy, throughput and cost efficiency expected to soon surpass srRNA-seq, and emergent sequencing platforms developing long-reads capabilities (for example, MGI's CycloneSEQ; representing an important addition to the field). However, precise control over RNA degradation is crucial to fully harness the benefits of capturing an entire RNA molecule in a single read. Hence, the technology must shift its focus to developing more efficient and robust library preparation protocols.

lrRNA-seq holds immense potential for the discovery of novel transcripts and contributes to genome annotation of biodiversity, but clear guidelines for the most effective experimental and bioinformatics pipelines have yet to be established. lrRNA-seq is also a valuable resource for expanding gene and transcript catalogues in well-studied organisms, and it can have a key role in pan-transcriptome projects that catalogue variants associated with specific populations, tissues or cell types. However, there is a risk of capturing spurious RNA molecules, leading to enormous, unwieldy RNA catalogues[74]. The genome annotation community must balance comprehensiveness and manageable data outputs. For quantification, the focus is shifting from LRS methods technologies generating sufficient reads for quantification to understanding the inherent biases in quantification and how these can be corrected. Length bias, which affects both transcript detection and quantification, is one such challenge that requires investigation. Moreover, the field still requires systematic studies on sequencing depth, replication, experimental design, as well as tailored methods for normalization and differential isoform usage analysis.

Emerging applications of lrRNA-seq represent opportunities to advance genome research, including full-length isoforms as biomarkers of disease or to better understand disease mechanisms, algorithms to profile RNA modifications beyond m^6A to fully include the epitranscriptome layer in transcriptome analysis, and integration with other sequencing and mass spectrometry methods for long-read multi-omics approaches that can better describe RNA regulatory dynamics[230]. Realizing these possibilities and developing new tools tailored to bulk, single-cell and spatial lrRNA-seq will require addressing the considerable computational demands of lrRNA-seq. Given the already high environmental cost of bioinformatics[231], optimizing data size and computational efficiency should be a top priority for the community.

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